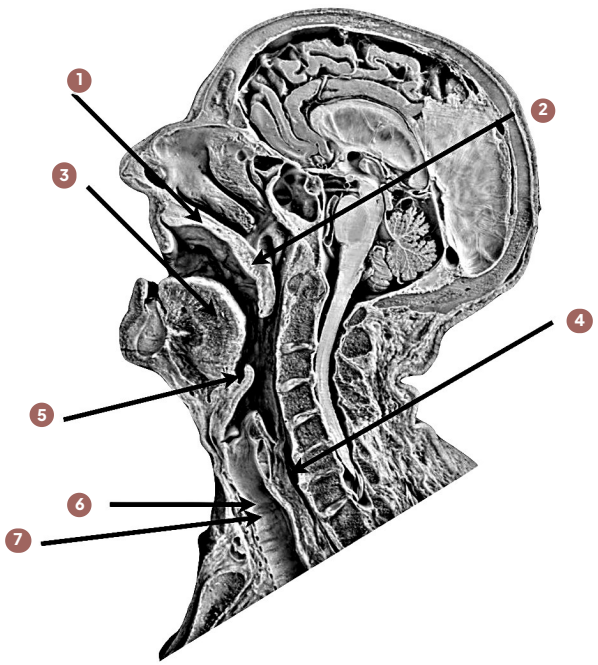
Respiratory System, Part 1

Competencies 97 to 99: airway map, nose and pharynx, larynx. Cadaver beside model. Fill in as we go.

Name _____

A. Nose, pharynx, larynx: the start of the airway (cadaver) COMPETENCY 97, 98



Cadaver, midsagittal head and neck

1. Hard palate
2. Soft palate
3. Tongue
4. Esophagus
5. Epiglottis
6. Vestibular folds (false vocal folds)
7. Vocal folds (true vocal cords)

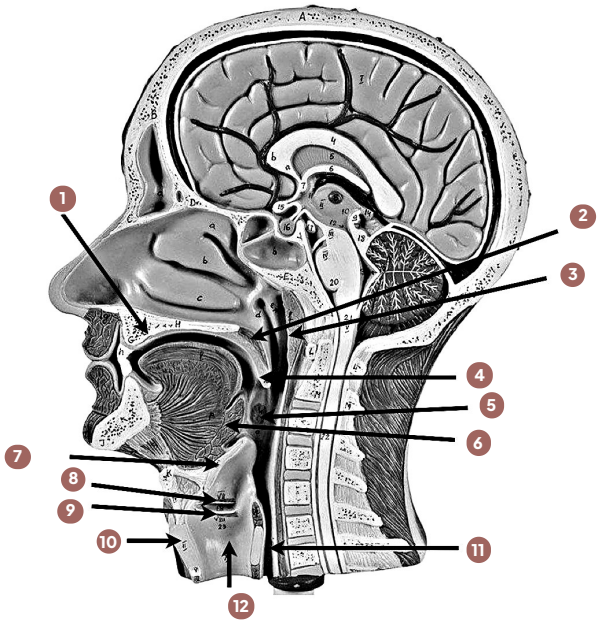
FILL IN

Air only region: nasopharynx

Air and food: oropharynx, laryngopharynx

Lower tract begins at the larynx

B. Palate, tonsils and the three regions of the pharynx (model) COMPETENCY 98, 113



Model, midsagittal head: palate, tonsils, larynx

1. Hard palate
2. Soft palate
3. Pharyngeal tonsil
4. Uvula
5. Palatine tonsil
6. Lingual tonsil
7. Epiglottis
8. Vestibular folds (false vocal folds)
9. Vocal folds (true vocal cords)
10. Thyroid cartilage
11. Esophagus
12. Larynx

PATHWAY OF ONE BREATH, IN ORDER

1. nares
2. nasal cavity
3. choanae
4. nasopharynx
5. oropharynx
6. laryngopharynx
7. larynx (glottis)
8. trachea

Extra notes and drawings

Extra notes and drawings

Respiratory System, Part 1

A. Nose, pharynx, larynx: the start of the airway (cadaver) · Competency 97, 98 · one card per numbered structure, same lines as the slide. Write the name, then fill each line as we go.

Name _____

1 Hard palate

Def Bony front part of the roof of the mouth
Contributing structures Palatine processes (maxillae); horizontal plates (palatine bones)
Borders Alveolar arches of the maxillae in front and on the sides; soft palate behind
Blood supply (info only) Greater palatine artery
Function Separates oral cavity from nasal cavity; the tongue presses food against it
Clinical Cleft palate when the two halves fail to fuse

2 Soft palate

Def Muscular back part of the roof of the mouth
Made of Skeletal muscle covered with mucosa; ends in the uvula
Borders Hard palate in front; free edge behind with the palatoglossal and palatopharyngeal arches on each side
Action Lifts during a swallow to close off the nasopharynx
Nerve Vagus (CN X) for all its muscles except tensor veli palatini (CN V3)
Clinical Vagus injury: the uvula deviates away from the damaged side; snoring and sleep apnea from a floppy soft palate

3 Tongue

Def Muscular organ on the floor of the mouth
Made of Skeletal muscle covered with mucosa
Parts Extrinsic muscles (genioglossus, hyoglossus, styloglossus, palatoglossus) move it; intrinsic muscles change its shape
Nerve Hypoglossal (CN XII) for all muscles except palatoglossus (vagus)
Blood supply (info only) Lingual artery
Clinical Hypoglossal injury: the protruded tongue deviates toward the damaged side

4 Esophagus

Def Collapsible muscular tube from the laryngopharynx to the stomach, about 25 cm (10 inches)
Location Behind the trachea, through the mediastinum and the esophageal hiatus at T10
Lining Stratified squamous epithelium; adventitia outside, above the diaphragm
Nerve Vagus (parasympathetic), sympathetic trunk
Blood supply (info only) Inferior thyroid, esophageal branches of the aorta, left gastric artery
Function Carries the bolus to the stomach by peristalsis
Clinical GERD when the lower esophageal sphincter fails; three narrowings where swallowed objects lodge

5 Epiglottis

Def Spoon-shaped flap of elastic cartilage at the top of the larynx
Attaches Inside the thyroid cartilage, below the laryngeal prominence
Action Tips over the laryngeal inlet during a swallow so food slides past into the esophagus
Nerve Sensation by the internal laryngeal branch of the vagus
Clinical Epiglottitis in a child can close the airway; never force the tongue down to look

6 Vestibular folds (false vocal folds)

Def Upper pair of folds inside the larynx
Function Help close the airway during a swallow
Note No part in making sound

7 Vocal folds (true vocal cords)

Def Lower pair of folds inside the larynx
Contributing structures Vocal ligament from each arytenoid cartilage to the thyroid cartilage, covered by thin mucosa
Nerve Recurrent laryngeal nerve moves them
Function Vibrate as air passes to make sound; the gap between them is the glottis
Clinical Paralysis of one fold gives hoarseness; overuse gives vocal nodules

Respiratory System, Part 1

B. Palate, tonsils and the three regions of the pharynx (model) · Competency 98, 113 · one card per numbered structure, same lines as the slide. Write the name, then fill each line as we go.

Name _____

1 Hard palate

Def Bony front part of the roof of the mouth
Contributing structures Palatine processes (maxillae); horizontal plates (palatine bones)
Borders Alveolar arches of the maxillae in front and on the sides; soft palate behind
Blood supply (info only) Greater palatine artery
Function Separates oral cavity from nasal cavity; the tongue presses food against it
Clinical Cleft palate when the two halves fail to fuse

2 Soft palate

Def Muscular back part of the roof of the mouth
Made of Skeletal muscle covered with mucosa; ends in the uvula
Borders Hard palate in front; free edge behind with the palatoglossal and palatopharyngeal arches on each side
Action Lifts during a swallow to close off the nasopharynx
Nerve Vagus (CN X) for all its muscles except tensor veli palatini (CN V3)
Clinical Vagus injury: the uvula deviates away from the damaged side; snoring and sleep apnea from a floppy soft palate

3 Pharyngeal tonsil

Def Single mass of lymphatic tissue
Location Back wall of the nasopharynx
Function Traps pathogens entering through the nose
Clinical Enlarged adenoids block the nose and the auditory tube, so children mouth-breathe and get ear infections

4 Uvula

Def Fleshy tag hanging from the back edge of the soft palate
Function Rises with the soft palate to seal the nasopharynx when you swallow

5 Palatine tonsil

Def Paired lymphatic tissue
Location Side walls of the oropharynx, between the palatoglossal and palatopharyngeal arches
Blood supply (info only) Tonsillar branch of the facial artery
Function Traps pathogens entering through the mouth
Clinical Tonsillitis; tonsillectomy bleeding comes from the tonsillar artery

6 Lingual tonsil

Def Lymphatic tissue on the root (back) of the tongue
Location Oropharynx, below the palatine tonsils

7 Epiglottis

Def Spoon-shaped flap of elastic cartilage at the top of the larynx
Attaches Inside the thyroid cartilage, below the laryngeal prominence
Action Tips over the laryngeal inlet during a swallow so food slides past into the esophagus
Nerve Sensation by the internal laryngeal branch of the vagus
Clinical Epiglottitis in a child can close the airway; never force the tongue down to look

8 Vestibular folds (false vocal folds)

Def Upper pair of folds inside the larynx
Function Help close the airway during a swallow
Note No part in making sound

9 Vocal folds (true vocal cords)

Def Lower pair of folds inside the larynx
Contributing structures Vocal ligament from each arytenoid cartilage to the thyroid cartilage, covered by thin mucosa
Nerve Recurrent laryngeal nerve moves them
Function Vibrate as air passes to make sound; the gap between them is the glottis
Clinical Paralysis of one fold gives hoarseness; overuse gives vocal nodules

10 Thyroid cartilage

Def Largest laryngeal cartilage, a shield of hyaline cartilage open at the back
Contributing structures Two plates (laminae) meeting in front at the laryngeal prominence
Function Protects the vocal folds; anchors the front end of the vocal ligaments

11 Esophagus

Def Collapsible muscular tube from the laryngopharynx to the stomach, about 25 cm (10 inches)
Location Behind the trachea, through the mediastinum and the esophageal hiatus at T10
Lining Stratified squamous epithelium; adventitia outside, above the diaphragm
Nerve Vagus (parasympathetic), sympathetic trunk
Blood supply (info only) Inferior thyroid, esophageal branches of the aorta, left gastric artery
Function Carries the bolus to the stomach by peristalsis
Clinical GERD when the lower esophageal sphincter fails; three narrowings where swallowed objects lodge

12 Larynx

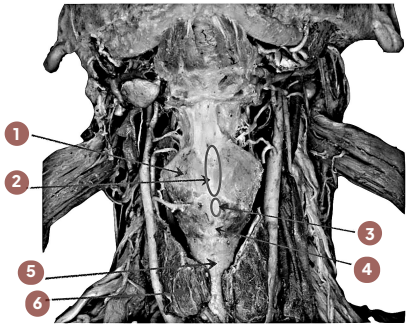
Def The voice box, a cartilage tube between the laryngopharynx and the trachea
Location C3 to C6, in front of the esophagus
Borders Hyoid bone above, trachea below, laryngopharynx behind
Parts Nine cartilages: thyroid, cricoid, epiglottis (single); arytenoid, corniculate, cuneiform (paired)
Nerve Vagus: recurrent laryngeal to all muscles but one, superior laryngeal to cricothyroid and sensation above the folds
Blood supply (info only) Superior and inferior laryngeal arteries
Function Keeps the airway open, keeps food out of it, makes the voice
Clinical Recurrent laryngeal nerve injury in thyroid surgery gives a hoarse voice

Respiratory System, Part 1

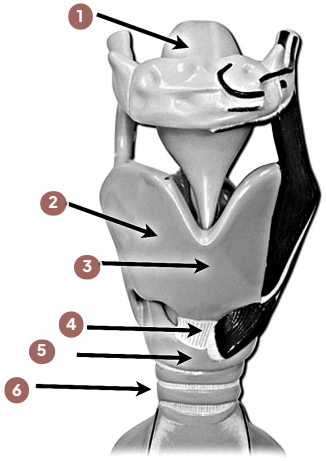
Competencies 97 to 99: airway map, nose and pharynx, larynx. Cadaver beside model. Fill in as we go.

Name _____

C. Larynx, anterior: cadaver beside model COMPETENCY 99



Cadaver, larynx and trachea, anterior



Model, larynx, anterior

CADAVER	MODEL
1 Thyroid cartilage	1 Epiglottis
2 Laryngeal prominence	2 Thyroid cartilage
3 Cricothyroid membrane	3 Laryngeal prominence
4 Cricoid cartilage	4 Cricothyroid membrane
5 Trachea	5 Cricoid cartilage
6 Thyroid gland	6 Trachea

FILL IN

Largest cartilage, the shield: thyroid

Its front ridge: laryngeal prominence

Only complete ring: cricoid

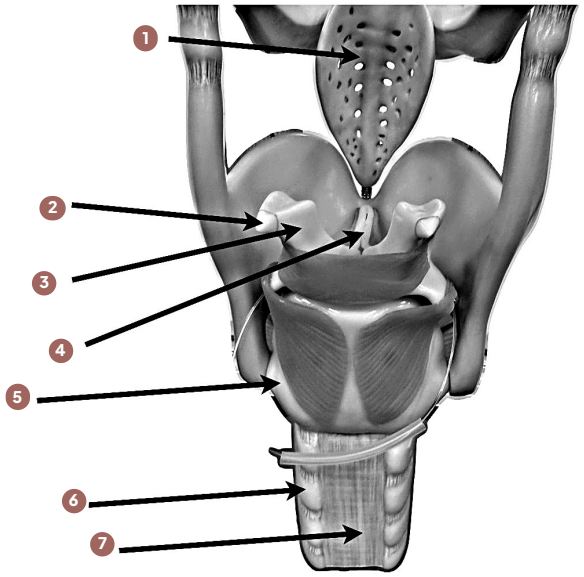
Emergency airway landmark: cricothyroid ligament

Single: thyroid, cricoid, epiglottis

Paired: arytenoid, corniculate, cuneiform

Extra notes and drawings

D. Larynx, posterior (model) COMPETENCY 99



Model, larynx, posterior

1. Epiglottis
2. Corniculate cartilage
3. Arytenoid cartilage
4. Vocal folds
5. Cricoid cartilage
6. Tracheal cartilages (C-rings)
7. Trachealis muscle

FILL IN

Elastic cartilage, not hyaline: epiglottis

Folds that make sound: vocal folds (true cords)

Folds that do not: vestibular folds (false cords)

Glottis = vocal folds plus the opening

Cartilage that pivots the folds: arytenoid

Open side of the C rings faces: back, toward the esophagus

Extra notes and drawings

Respiratory System, Part 1

C. Larynx, anterior: cadaver beside model · Competency 99 · one card per numbered structure, same lines as the slide. Write the name, then fill each line as we go.

Name _____

1 Thyroid cartilage

Def Largest laryngeal cartilage, a shield of hyaline cartilage open at the back

Contributing structures Two plates (laminae) meeting in front at the laryngeal prominence

Function Protects the vocal folds; anchors the front end of the vocal ligaments

2 Laryngeal prominence

Def The Adam's apple; front ridge where the two plates of the thyroid cartilage meet

Note Larger in males after puberty

3 Cricothyroid membrane

Def Membrane between the thyroid and cricoid cartilages in front

Location Just under the skin, below the vocal folds

Clinical Site of an emergency airway (cricothyrotomy) when the larynx above is blocked

4 Cricoid cartilage

Def Signet-ring cartilage below the thyroid cartilage, narrow in front and wide behind

Made of Hyaline cartilage

Note The only complete ring in the airway; the trachea hangs from its lower edge; level with C6

Clinical Cricoid pressure during intubation presses the esophagus shut against the spine

5 Trachea

Def The windpipe, from the cricoid cartilage into the mediastinum

Location In front of the esophagus, 10 to 12 cm long; forks at the sternal angle, T4 to T5

Made of 16 to 20 C-shaped rings of hyaline cartilage, trachealis muscle, respiratory epithelium

Blood supply (info only) Inferior thyroid and bronchial arteries

Function Carries air between the larynx and the primary bronchi

Clinical Tracheostomy below the isthmus of the thyroid; a goiter can press on it

6 Thyroid gland

Def Butterfly-shaped endocrine gland

Location Across the front of the trachea, just below the larynx, rings 2 to 4

Parts Two lobes joined by an isthmus

Blood supply (info only) Superior thyroid (external carotid) and inferior thyroid (thyrocervical trunk) arteries

Clinical Goiter can compress the trachea; the recurrent laryngeal nerves run right behind it and are at risk in surgery

1 Epiglottis

Def Spoon-shaped flap of elastic cartilage at the top of the larynx

Attaches Inside the thyroid cartilage, below the laryngeal prominence

Action Tips over the laryngeal inlet during a swallow so food slides past into the esophagus

Nerve Sensation by the internal laryngeal branch of the vagus

Clinical Epiglottitis in a child can close the airway; never force the tongue down to look

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Blood supply (info only) Inferior thyroid and bronchial arteries

Function Carries air between the larynx and the primary bronchi

Clinical Tracheostomy below the isthmus of the thyroid; a goiter can press on it

Respiratory System, Part 1

D. Larynx, posterior (model) · Competency 99 · one card per numbered structure, same lines as the slide. Write the name, then fill each line as we go.

Name _____

1 Epiglottis

Def Spoon-shaped flap of elastic cartilage at the top of the larynx

Attaches Inside the thyroid cartilage, below the laryngeal prominence

Action Tips over the laryngeal inlet during a swallow so food slides past into the esophagus

Nerve Sensation by the internal laryngeal branch of the vagus

Clinical Epiglottitis in a child can close the airway; never force the tongue down to look

2 Corniculate cartilage

Def Small paired horns of elastic cartilage

Location On the tips of the arytenoid cartilages

3 Arytenoid cartilage

Def Paired pyramid-shaped cartilages sitting on the back of the cricoid

Function Anchor the back end of the vocal folds; pivot to open and close the glottis

Note The most important paired cartilages

4 Vocal folds

Def Lower pair of folds inside the larynx, the true vocal cords

Contributing structures Vocal ligament from each arytenoid cartilage to the thyroid cartilage

Nerve Recurrent laryngeal nerve

Function Vibrate as air passes to make sound

Clinical Paralysis of one fold gives hoarseness

5 Cricoid cartilage

Def Signet-ring cartilage below the thyroid cartilage, narrow in front and wide behind

Made of Hyaline cartilage

Note The only complete ring in the airway; the trachea hangs from its lower edge; level with C6

Clinical Cricoid pressure during intubation presses the esophagus shut against the spine

6 Tracheal cartilages (C-rings)

Def 16 to 20 C-shaped rings of hyaline cartilage in the tracheal wall

Note The open side of each C faces back toward the esophagus

Function Hold the trachea open at all times

7 Trachealis muscle

Def Smooth muscle bridging the open back of each C ring

Location Posterior wall of the trachea, against the esophagus

Action Narrows the trachea during a cough; lets the esophagus bulge forward during a swallow

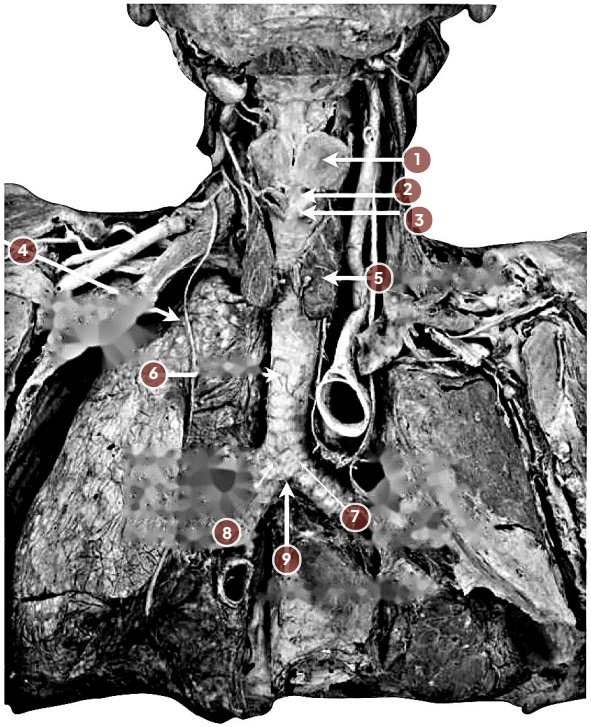
Nerve Vagus nerve, parasympathetic

Respiratory System, Part 2

Competencies 100 and 101: trachea, bronchial tree, lungs and pleurae. Cadaver beside model. Fill in as we go.

Name _____

A. Trachea, carina and main bronchi (cadaver) COMPETENCY 100, 103



Cadaver, neck and thorax, anterior

1. Thyroid cartilage
2. Cricothyroid membrane
3. Cricoid cartilage
4. Phrenic nerve
5. Thyroid gland
6. Trachea
7. Left primary bronchus
8. Right primary bronchus
9. Carina

FILL IN

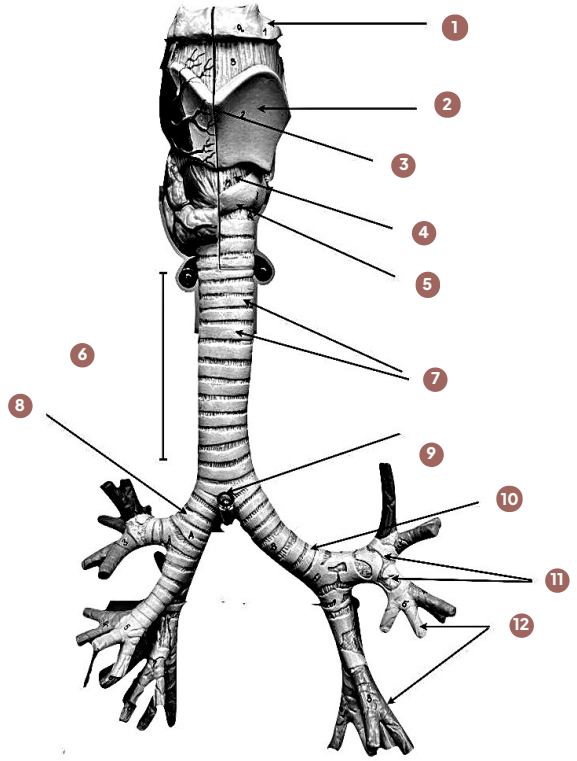
Carina sits at: the sternal angle, T4 to T5

Right main bronchus is: wider, shorter, more vertical

so an inhaled object lands: in the right lung

Nerve to the diaphragm: phrenic, C3 to C5

B. The bronchial tree, trunk to twigs (model) COMPETENCY 100



Model, larynx, trachea and bronchial tree

1. Hyoid bone
2. Thyroid cartilage
3. Laryngeal prominence
4. Cricothyroid ligament
5. Cricoid cartilage
6. Trachea
7. Tracheal cartilages (C-rings)
8. Right primary bronchus
9. Carina
10. Left primary bronchus
11. Secondary (lobar) bronchi
12. Tertiary (segmental) bronchi

DOWN THE TREE, IN ORDER

1. trachea
2. main (primary) bronchi
3. lobar (secondary) bronchi
4. segmental (tertiary) bronchi
5. bronchioles
6. terminal bronchioles

FILL IN

Cartilage: rings, then: plates, then none

Smooth muscle: increases

Extra notes and drawings

Extra notes and drawings

Respiratory System, Part 2

A. Trachea, carina and main bronchi (cadaver) · Competency 100, 103 · one card per numbered structure, same lines as the slide. Write the name, then fill each line as we go.

Name _____

1 Thyroid cartilage

Def Largest laryngeal cartilage, a shield of hyaline cartilage open at the back

Contributing structures Two plates (laminae) meeting in front at the laryngeal prominence

Function Protects the vocal folds; anchors the front end of the vocal ligaments

2 Cricothyroid membrane

Def Membrane between the thyroid and cricoid cartilages in front

Location Just under the skin, below the vocal folds

Clinical Site of an emergency airway (cricothyrotomy) when the larynx above is blocked

3 Cricoid cartilage

Def Signet-ring cartilage below the thyroid cartilage, narrow in front and wide behind

Made of Hyaline cartilage

Note The only complete ring in the airway; the trachea hangs from its lower edge; level with C6

Clinical Cricoid pressure during intubation presses the esophagus shut against the spine

4 Phrenic nerve

Def The nerve to the diaphragm

Origin C3, C4, C5

Course Down the side of the pericardium, in front of the root of the lung

Supplies Motor to the diaphragm; sensation from the pericardium, mediastinal pleura, and central diaphragm

Clinical Spinal injury above C3 stops breathing; diaphragm irritation refers pain to the shoulder (C3 to C5 dermatomes)

5 Thyroid gland

Def Butterfly-shaped endocrine gland

Location Across the front of the trachea, just below the larynx, rings 2 to 4

Parts Two lobes joined by an isthmus

Blood supply (info only) Superior thyroid (external carotid) and inferior thyroid (thyrocervical trunk) arteries

Clinical Goiter can compress the trachea; the recurrent laryngeal nerves run right behind it and are at risk in surgery

6 Trachea

Def The windpipe, from the cricoid cartilage into the mediastinum

Location In front of the esophagus, 10 to 12 cm long; forks at the sternal angle, T4 to T5

Made of 16 to 20 C-shaped rings of hyaline cartilage, trachealis muscle, respiratory epithelium

Blood supply (info only) Inferior thyroid and bronchial arteries

Function Carries air between the larynx and the primary bronchi

Clinical Tracheostomy below the isthmus of the thyroid; a goiter can press on it

7 Left primary bronchus

Def Left branch of the trachea

Note Narrower, longer, and more horizontal because it angles under the aortic arch and around the heart

Blood supply (info only) Bronchial arteries from the aorta

8 Right primary bronchus

Def Right branch of the trachea

Note Wider, shorter, and more vertical than the left

Blood supply (info only) Bronchial arteries from the aorta

Clinical Inhaled objects usually drop into the right lung, most often the lower lobe

9 Carina

Def Internal ridge where the trachea forks into the two primary bronchi

Location At the sternal angle, T4 to T5

Made of The lowest tracheal cartilage

Nerve Vagus; densely supplied with sensory endings

Clinical Touching it with a suction catheter or inhaled object sets off a violent cough

Respiratory System, Part 2

B. The bronchial tree, trunk to twigs (model) · Competency 100 · one card per numbered structure, same lines as the slide. Write the name, then fill each line as we go.

Name _____

1 Hyoid bone
Def U-shaped bone in the neck above the larynx, at C3
Attaches Suprahyoid muscles above, infrahyoid muscles below; the larynx hangs from it by the thyrohyoid membrane
Note Not part of the larynx; the only bone with no joint to another bone
Clinical A fractured hyoid is a classic sign of strangulation

2 Thyroid cartilage
Def Largest laryngeal cartilage, a shield of hyaline cartilage open at the back
Contributing structures Two plates (laminae) meeting in front at the laryngeal prominence
Function Protects the vocal folds; anchors the front end of the vocal ligaments

3 Laryngeal prominence
Def The Adam's apple; front ridge where the two plates of the thyroid cartilage meet
Note Larger in males after puberty

4 Cricothyroid ligament
Def Membrane filling the gap between the thyroid and cricoid cartilages in front
Location Just under the skin, below the vocal folds
Clinical Site of an emergency airway (cricothyrotomy) when the larynx above is blocked

5 Cricoid cartilage
Def Signet-ring cartilage below the thyroid cartilage, narrow in front and wide behind
Made of Hyaline cartilage
Note The only complete ring in the airway; the trachea hangs from its lower edge; level with C6
Clinical Cricoid pressure during intubation presses the esophagus shut against the spine

6 Trachea
Def The windpipe, from the cricoid cartilage into the mediastinum
Location In front of the esophagus, 10 to 12 cm long; forks at the sternal angle, T4 to T5
Made of 16 to 20 C-shaped rings of hyaline cartilage, trachealis muscle, respiratory epithelium
Blood supply (info only) Inferior thyroid and bronchial arteries
Function Carries air between the larynx and the primary bronchi
Clinical Tracheostomy below the isthmus of the thyroid; a goiter can press on it

7 Tracheal cartilages (C-rings)
Def 16 to 20 C-shaped rings of hyaline cartilage in the tracheal wall
Note The open side of each C faces back toward the esophagus
Function Hold the trachea open at all times

8 Right primary bronchus
Def Right branch of the trachea
Note Wider, shorter, and more vertical than the left
Blood supply (info only) Bronchial arteries from the aorta
Clinical Inhaled objects usually drop into the right lung, most often the lower lobe

9 Carina
Def Internal ridge where the trachea forks into the two primary bronchi
Location At the sternal angle, T4 to T5
Made of The lowest tracheal cartilage
Nerve Vagus; densely supplied with sensory endings
Clinical Touching it with a suction catheter or inhaled object sets off a violent cough

10 Left primary bronchus
Def Left branch of the trachea
Note Narrower, longer, and more horizontal because it angles under the aortic arch and around the heart
Blood supply (info only) Bronchial arteries from the aorta

11 Secondary (lobar) bronchi
Def Branches of each primary bronchus, one per lobe
Note Three on the right, two on the left

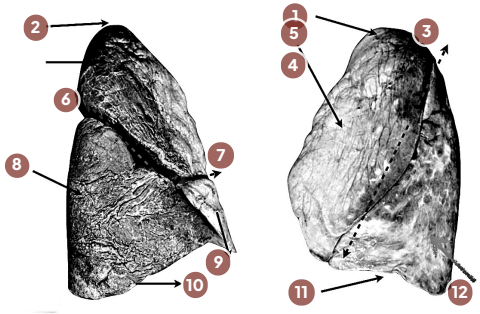
12 Tertiary (segmental) bronchi
Def Branches of the lobar bronchi, one per bronchopulmonary segment
Note Ten on the right, eight to ten on the left

Respiratory System, Part 2

Competencies 100 and 101: trachea, bronchial tree, lungs and pleurae. Cadaver beside model. Fill in as we go.

Name _____

C. Lobes and fissures (cadaver) COMPETENCY 101



Cadaver, right and left lungs, anterior

1. Apex
2. Apex
3. Oblique fissure
4. Superior lobe
5. Superior lobe
6. Oblique fissure
7. Horizontal fissure
8. Inferior lobe
9. Middle lobe
10. Base
11. Base
12. Inferior lobe

FILL IN

Right lung lobes: superior, middle, inferior

Right fissures: horizontal and oblique

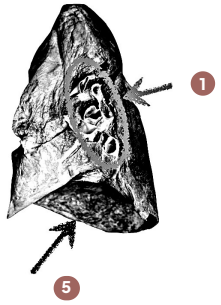
Left lung lobes: superior, inferior

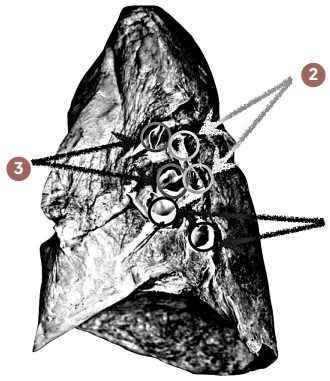
Left fissure: oblique only

Left lung only: cardiac notch, lingula

Extra notes and drawings

D. Hilum and pleurae (cadaver, and the two-layer rule) COMPETENCY 101





Cadaver, right lung, medial view

ary arteries

ary veins

gmatic surface

layer

layer

cavity

led)

4 Pleural cavity

5 Parietal pleura

6 Visceral pleura

es through the hilum

arteries carry oxygen-poor blood in

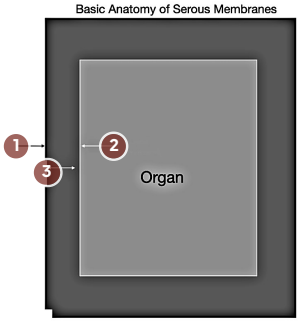
veins carry oxygen-rich blood out

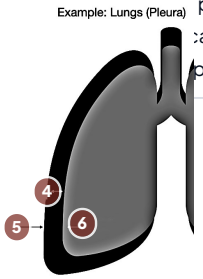
Parietal pleura lines the chest wall

pleura covers the lung

cavity holds a film of fluid

pocket: costodiaphragmatic recess





Serous membrane: parietal layer, visceral layer, cavity between

Extra notes and drawings

Respiratory System, Part 2

C. Lobes and fissures (cadaver) · Competency 101 · one card per numbered structure, same lines as the slide. Write the name, then fill each line as we go.

Name _____

1 Apex
Def Narrow top of the lung
Location Rises just above the clavicle into the root of the neck

2 Apex
Def Narrow top of the lung
Location Rises just above the clavicle into the root of the neck

3 Oblique fissure
Def Deep groove on both lungs
Function Separates the inferior lobe from the lobe or lobes above it

4 Superior lobe
Def Top lobe of each lung
Note On the left it carries the cardiac notch and the lingula

5 Superior lobe
Def Top lobe of each lung
Note On the left it carries the cardiac notch and the lingula

6 Oblique fissure
Def Deep groove on both lungs
Function Separates the inferior lobe from the lobe or lobes above it

7 Horizontal fissure
Def Groove found only on the right lung
Function Separates the superior lobe from the middle lobe

8 Inferior lobe
Def Lowest lobe of each lung
Location Below the oblique fissure

9 Middle lobe
Def Lobe found only on the right lung
Location Between the horizontal and oblique fissures
Note The left lung has no middle lobe; the heart takes that room

10 Base
Def Broad concave bottom of the lung
Location Rests on the dome of the diaphragm

11 Base
Def Broad concave bottom of the lung
Location Rests on the dome of the diaphragm

12 Inferior lobe
Def Lowest lobe of each lung
Location Below the oblique fissure

Respiratory System, Part 2

D. Hilum and pleurae (cadaver, and the two-layer rule) · Competency 101 · one card per numbered structure, same lines as the slide. Write the name, then fill each line as we go.

Name _____

1 Hilum

Def Indented area on the medial surface of the lung

Contributing structures Primary bronchus, pulmonary artery, two pulmonary veins, bronchial vessels, nerves, lymphatics (the root of the lung)

Borders Pulmonary ligament hangs from its lower edge

Note Right: bronchus highest. Left: artery highest. Veins lowest on both sides

2 Bronchi

Def The branching airways inside the lung, from primary to tertiary

Location Enter the lung at the hilum

Note Right hilum: bronchus at or above the artery. Left hilum: artery is highest

3 Pulmonary arteries

Def Right and left arteries from the pulmonary trunk

Carries Oxygen-poor blood from the right ventricle into the lung

Note In the left hilum the artery is the highest structure

4 Pulmonary veins

Def Two veins from each lung to the left atrium

Carries Oxygen-rich blood back to the heart

Note Lowest structures in both hila

5 Diaphragmatic surface

Def The base of the lung

Location Rests on the dome of the diaphragm

1 Parietal layer

Def Layer of a serous membrane that lines the wall of the cavity

2 Visceral layer

Def Layer of a serous membrane that covers the organ itself

3 Serous cavity (fluid-filled)

Def Thin space between the parietal and visceral layers

Contents A film of serous fluid

Function Lets the organ glide against the wall

4 Pleural cavity

Def Thin space between the parietal and visceral pleurae

Contents A film of serous fluid, nothing else

Function Lets the lung slide without friction; the fluid also holds the lung to the chest wall

Clinical Pneumothorax: air in the cavity lets the lung collapse. Pleural effusion: fluid collects. Each side is separate, so one lung can collapse alone

5 Parietal pleura

Def Outer layer of the serous membrane around the lung

Location Lines the thoracic wall, the top of the diaphragm, and the sides of the mediastinum

Nerve (info only) Intercostal and phrenic nerves, so it feels pain

Clinical Pleurisy pain comes from this layer; a thoracentesis needle passes through it

6 Visceral pleura

Def Inner layer of the serous membrane around the lung

Location Clings to the lung surface and dips into the fissures

Nerve (info only) Autonomic only; no pain sensation

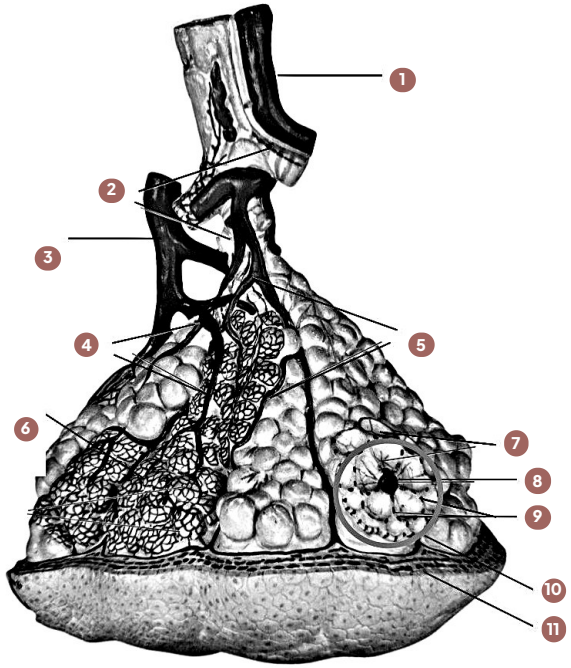
Note Continuous with the parietal pleura at the hilum

Respiratory System, Part 3

Competencies 102 to 104, with 97, 100 and 103: respiratory zone, histology, thoracic wall and diaphragm, disorders.

Name _____

A. Respiratory zone and the alveolus (model) COMPETENCY 97, 102



Model, lung lobule

1. Branches of pulmonary vein
2. Bronchioles
3. Branches of pulmonary artery
4. Pulmonary arterioles
5. Pulmonary venules
6. Capillaries surrounding alveoli
7. Alveolus
8. Opening of the alveolar duct
9. Pulmonary capillaries
10. Alveolar duct
11. Visceral pleura

FILL IN

Conducting zone ends at the: terminal bronchiole

Respiratory zone begins at the: respiratory bronchiole

Type I cell: thin and flat, gas crosses

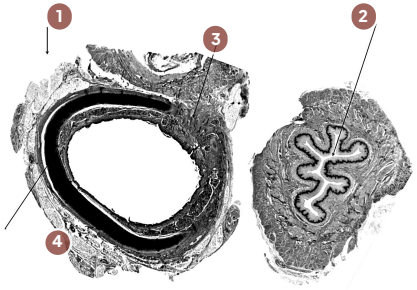
Type II cell: cuboidal, makes surfactant

Macrophage (dust cell): cleans up

Membrane, air to blood: type I cell, fused basement membranes, endothelium

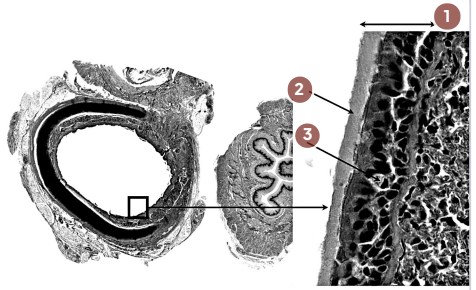
Extra notes and drawings

B. Histology: trachea and respiratory epithelium COMPETENCY 100, 102



Slide, trachea and esophagus, cross section

1. Trachea
2. Esophagus
3. Trachealis muscle
4. Tracheal cartilage (hyaline cartilage)



Slide, respiratory epithelium

1. Pseudostratified ciliated columnar epithelium
2. Cilia
3. Goblet cell

FILL IN

Open side of each C ring faces the: esophagus

Trachea epithelium: pseudostratified ciliated columnar

Goblet cells make: mucus

Cilia sweep it toward the: pharynx

Alveolar wall epithelium: simple squamous

Extra notes and drawings

Respiratory System, Part 3

A. Respiratory zone and the alveolus (model) · Competency 97, 102 · one card per numbered structure, same lines as the slide. Write the name, then fill each line as we go.

Name _____

1 Branches of pulmonary vein

Def Tributaries of the pulmonary veins inside the lung

Carries Oxygen-rich blood from the alveolar capillaries toward the left atrium

2 Bronchioles

Def Airways under 1 mm across, past the tertiary bronchi

Made of Smooth muscle and simple epithelium; no cartilage

Nerve Parasympathetic (vagus) constricts; sympathetic dilates

Clinical Asthma: the smooth muscle narrows them with nothing to hold them open; albuterol relaxes it

3 Branches of pulmonary artery

Def Branches of the pulmonary artery inside the lung

Carries Oxygen-poor blood from the right ventricle to the alveolar capillaries

4 Pulmonary arterioles

Def Smallest branches of the pulmonary artery

Function Feed the capillary net around the alveoli

5 Pulmonary venules

Def Smallest tributaries of the pulmonary veins

Function Drain the alveolar capillaries

6 Capillaries surrounding alveoli

Def Dense net of capillaries wrapped around each alveolus

Carries Oxygen-poor blood in from the pulmonary arteries, oxygen-rich blood out to the pulmonary veins

Note Where they touch the alveolar wall the respiratory membrane is three layers thick

7 Alveolus

Def Single thin-walled air sac at the end of the airway

Made of Type I cells (thin wall), type II cells (surfactant), alveolar macrophages

Blood supply (info only) Pulmonary capillary net on its outer surface

Function The site where oxygen and carbon dioxide cross between air and blood

Clinical Emphysema destroys alveolar walls; premature infants lack surfactant

8 Opening of the alveolar duct

Def Where an alveolar duct opens into a cluster of alveoli, the alveolar sac

9 Pulmonary capillaries

Def Capillaries wrapped around each alveolus

Function Gas exchange across the three-layer respiratory membrane

10 Alveolar duct

Def Passage whose walls are made entirely of alveoli

Location Respiratory zone, after the respiratory bronchiole

Function Leads air into the alveolar sacs

11 Visceral pleura

Def Inner layer of the serous membrane around the lung

Location Clings to the lung surface and dips into the fissures

Nerve (info only) Autonomic only; no pain sensation

Note Continuous with the parietal pleura at the hilum

Respiratory System, Part 3

B. Histology: trachea and respiratory epithelium · Competency 100, 102 · one card per numbered structure, same lines as the slide. Write the name, then fill each line as we go.

Name _____

1 Trachea

Def The windpipe, from the cricoid cartilage into the mediastinum

Location In front of the esophagus, 10 to 12 cm long; forks at the sternal angle, T4 to T5

Made of 16 to 20 C-shaped rings of hyaline cartilage, trachealis muscle, respiratory epithelium

Blood supply (info only) Inferior thyroid and bronchial arteries

Function Carries air between the larynx and the primary bronchi

Clinical Tracheostomy below the isthmus of the thyroid; a goiter can press on it

2 Esophagus

Def Collapsible muscular tube from the laryngopharynx to the stomach, about 25 cm (10 inches)

Location Behind the trachea, through the mediastinum and the esophageal hiatus at T10

Lining Stratified squamous epithelium; adventitia outside, above the diaphragm

Nerve Vagus (parasympathetic), sympathetic trunk

Blood supply (info only) Inferior thyroid, esophageal branches of the aorta, left gastric artery

Function Carries the bolus to the stomach by peristalsis

Clinical GERD when the lower esophageal sphincter fails; three narrowings where swallowed objects lodge

3 Trachealis muscle

Def Smooth muscle bridging the open back of each C ring

Location Posterior wall of the trachea, against the esophagus

Action Narrows the trachea during a cough; lets the esophagus bulge forward during a swallow

Nerve Vagus nerve, parasympathetic

4 Tracheal cartilage (hyaline cartilage)

Def One of the C-shaped rings in the tracheal wall

Made of Hyaline cartilage

Function Keeps the trachea from collapsing; the open back is bridged by the trachealis muscle

1 Pseudostratified ciliated columnar epithelium

Def The respiratory epithelium

Made of Tall ciliated cells and goblet cells on one basement membrane, nuclei at different heights

Location Nasal cavity, trachea, bronchi

Function Mucus traps debris; cilia sweep it toward the pharynx

2 Cilia

Def Hairlike projections on the respiratory epithelium

Function Beat toward the pharynx to move mucus, dust, and microbes up and out

3 Goblet cell

Def Mucus-secreting cell scattered in the respiratory epithelium

Function Its mucus traps dust and microbes for the cilia to clear

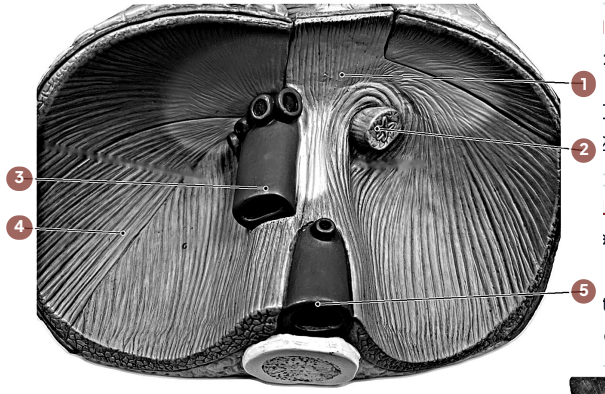
Respiratory System, Part 3

Competencies 102 to 104, with 97, 100 and 103: respiratory zone, histology, thoracic wall and diaphragm, disorders.

Name _____

C. Thoracic wall and diaphragm: the three openings

COMPETENCY 103



Model, diaphragm from above

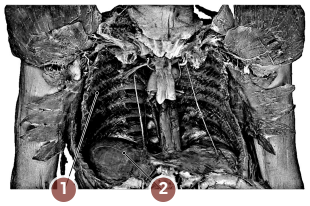
1. Central tendon
2. Esophagus
3. Inferior vena cava
4. Diaphragm
5. Aorta

ROUGH THE DIAPHRAGM, TOP TO BOTTOM

- : inferior vena cava, caval opening
- 1: esophagus, esophageal hiatus, with the vagus nerves
- 2: aorta, aortic hiatus, with the thoracic duct

L IN

- ain muscle of breathing: diaphragm
- nerve and levels: phrenic nerve, C3, C4, C5
- ternal intercostal fibers run down and forward
- ernal intercostals work in forced expiration



Cadaver, thorax opened: the diaphragm below the lungs

1. Innermost intercostals
2. Diaphragm

Extra notes and drawings

D. Disorders: which structure does each one affect?

COMPETENCY 104

DISORDER: STRUCTURE, WHAT CHANGES

- Asthma: bronchi and bronchioles narrow, no cartilage to hold them open
- Emphysema: alveolar walls destroyed, permanent
- Pneumonia: alveoli fill with fluid
- Pneumothorax: air in the pleural cavity, lung collapses
- Pleural effusion: fluid in the pleural cavity
- COPD: emphysema plus chronic bronchitis
- Infant RDS: type II cells, not enough surfactant
- Lung cancer: lung tissue, tumor from airway epithelium

DRAW: ONE LUNG IN ITS TWO PLEURAL LAYERS, THEN SHOW A PNEUMOTHORAX AND AN EFFUSION ON IT

Extra notes and drawings

Respiratory System, Part 3

C. Thoracic wall and diaphragm: the three openings · Competency 103 · one card per numbered structure, same lines as the slide. Write the name, then fill each line as we go.

Name _____

1 Central tendon

Def Flat sheet of tendon at the top of the diaphragm's dome
Function All the diaphragm's fibers pull toward it
Note The inferior vena cava passes through it at T8

2 Esophagus

Def Collapsible muscular tube from the laryngopharynx to the stomach, about 25 cm (10 inches)
Location Behind the trachea, through the mediastinum and the esophageal hiatus at T10
Lining Stratified squamous epithelium; adventitia outside, above the diaphragm
Nerve Vagus (parasympathetic), sympathetic trunk
Blood supply (info only) Inferior thyroid, esophageal branches of the aorta, left gastric artery
Function Carries the bolus to the stomach by peristalsis
Clinical GERD when the lower esophageal sphincter fails; three narrowings where swallowed objects lodge

3 Inferior vena cava

Def Largest vein of the lower body
Course Forms at L5 from the two common iliac veins, runs up the right side of the aorta, through the central tendon at T8
Carries Oxygen-poor blood from everything below the diaphragm to the right atrium
Clinical Compressed by the uterus late in pregnancy when the patient lies on her back

4 Diaphragm

Def Dome-shaped sheet of skeletal muscle that floors the thorax
Origin Xiphoid process, inner surfaces of the lower six ribs, and the crura on L1 to L3
Insertion Central tendon
Action Contracts and flattens to make the thorax taller: inspiration
Nerve (info only) Phrenic nerves, C3 to C5
Blood supply (info only) Pericardiacophrenic, musculophrenic, and inferior phrenic arteries
Openings Caval hiatus T8 (IVC), esophageal hiatus T10, aortic hiatus T12
Clinical Hiatal hernia through the esophageal hiatus; a phrenic nerve injury paralyzes that half of the dome

5 Aorta

Def Largest artery, from the left ventricle
Course Behind the diaphragm through the aortic hiatus at T12, between the crura, with the thoracic duct
Parts Thoracic aorta above the diaphragm, abdominal aorta below it
Clinical Abdominal aortic aneurysm, usually below the renal arteries

1 Innermost intercostals

Def Deepest of the three intercostal layers
Location Inside the internal intercostals, separated from them by the intercostal vessels and nerve
Action Assist the internal intercostals in forced expiration
Nerve (info only) Intercostal nerves
Clinical A chest tube is placed along the top of a rib to miss the vein, artery, and nerve that run under the rib above

2 Diaphragm

Def Dome-shaped sheet of skeletal muscle that floors the thorax
Origin Xiphoid process, inner surfaces of the lower six ribs, and the crura on L1 to L3
Insertion Central tendon
Action Contracts and flattens to make the thorax taller: inspiration
Nerve (info only) Phrenic nerves, C3 to C5
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Openings Caval hiatus T8 (IVC), esophageal hiatus T10, aortic hiatus T12
Clinical Hiatal hernia through the esophageal hiatus; a phrenic nerve injury paralyzes that half of the dome